Mohammed Chakib Belgaid, Ph.D.

Applied AI & Product Engineer · Ph.D. in Software Engineering

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PROFILE

Applied AI and product engineer with a Ph.D. in software engineering and 10+ years across backend platforms, developer tools, research infrastructure, and startup delivery. Builds measurable systems end to end—from product workflow and APIs through local/open-model evaluation, cloud deployment, and systems-level instrumentation.

EXPERIENCE

AI Research Engineer / Lead AI Engineer — MCQ Scan 2024–present · Lille, France

- Led a production-facing no-code multi-agent platform from API architecture and frontend integration through deployment.
- Reduced end-to-end workflow processing time by 60% through a parallelizable dependency graph, routing simpler steps to smaller models, and caching repeated calls (internal measurements).
- Fine-tuned Llama 3.1 8B with LoRA. Internal evaluation: 97% of requests structurally valid and enum-compliant; 95% semantic accuracy per attribute.
- Evaluated constrained-decoding tools and integrated LM Format Enforcer with vLLM to enforce allowed enum values, alongside Pydantic validation for business coherence.
- Built secure AI deployment paths on GCP with Cloud Run, Secret Manager, Artifact Registry, Docker, and GitHub Actions.
- Conceived AgentDiet, built its Biscuit authorization demonstrator, and prepared the technical proposal, market analysis, budget and plan; awarded a 2026 Hyper Open X grant funded by France 2030.

Research Engineer — Inria, ANR project with Qarnot Computing 2022–2024 · France

- Adapted the existing HWPC sensor to the infrastructure of Qarnot Computing, the industrial partner of the ANR project.
- Translated research tooling into operational infrastructure with explicit reproducibility and measurement-reliability constraints.

Ph.D. Researcher — University of Lille / Inria 2018–2022 · Lille, France

- Conducted empirical research on green coding and energy-aware software using a test–measure–optimize approach; thesis defended in December 2022.

Co-founder / CTO — Funecs 2014–2018 · Algeria

- Led product, architecture, team coordination, and client delivery for a serious-games advertising startup.
- Won Ooredoo T-Start in 2014 and delivered AquaRings and Get the 7 in Unity. The 2016 national campaign attracted over 300,000 unique subscriber numbers; led six employees and two interns.

SELECTED PROJECTS

BP Log / bp-tracker

Offline browser OCR for monitor displays: synthetic-data training, a 10,123-parameter CNN, and plain-JavaScript inference. 91% full readings on 55 development photos; 99.5% digit accuracy on 187 held-out digits from one monitor. Source

Whispbook

Self-hosted React/FastAPI audiobook studio with local TTS, document editing, character voices, chapter audio, subtitles, M4B export, and reproducible generation scripts. Separate from the speaker-attribution experiments. Source

Wattch Core

Measurement daemon and clients with Linux RAPL, a versioned Unix-socket protocol, replayable traces, and Python/VS Code tools. Rust implementation developed with AI assistance; Rust reading proficiency, Python expertise. Source

pyJoules

Developed and maintained pyJoules and co-developed pyRAPL for measuring software energy use; created JouleHunter for Python energy profiling. Source

SKILLS

Product & frontend: React, TypeScript, workflow design, frontend integration

Backend & cloud: Python, FastAPI, Django, Docker, GCP, Cloud Run, GitHub Actions

Applied AI: agent workflows, tool calling, structured outputs, local LLMs, LoRA, Hugging Face

Systems & measurement: Linux, RAPL, PowerAPI, Unix sockets; Rust code reading (AI-assisted implementation)

Languages: French (bilingual)

EDUCATION

Ph.D., Software Engineering / Energy-Aware Software — University of Lille / Inria, 2022

Engineering Degree, Computer Science — École Nationale Supérieure d'Informatique, Algiers, 2017

Public portfolio edition. Direct contact and profile links are included; phone number is intentionally omitted.